Flight Dynamics of the MBB Bo 105 in JSBSim

An elastic-blade hingeless rotor model, checked against DLR and MBB flight tests and NASA's full-scale rotor tests

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Abstract

A flight-dynamics model of the MBB Bo 105 for the FlightGear simulator has been written in JSBSim, replacing the parametric YASim model the aircraft used before. The Bo 105's hingeless, soft in-plane rotor is what sets it apart, and the model is built around it: the four blades are simulated individually as elastic beams cantilevered from the hub with 2.5° of precone, each with its first flap and lag bending modes, computed from the blade's mass and stiffness distribution, and a quasi-static first torsion mode; eight segments per blade use the NACA 23012 tables of Boeing Vertol's C-81 analysis at their own angle of attack and Mach number, under a Peters–He finite-state dynamic wake. The hub moments come from the blades' root bending, the rotor speed and the four lag accelerations are solved together, and a ring-vortex wake carries the rotor's downwash to the fuselage, stabiliser, end plates and tail rotor. A blade-element teetering tail rotor and two Allison 250-C20B engines with their governors complete it. The whole model, 985 aerodynamic functions and 405 control-system components, is generated by a script from transcribed data and runs in plain JSBSim at 960 Hz without scripting. It was flown offline through a scripted programme. Level-flight trims agree with MBB's flight test within 1.0° of collective, 1.2° of longitudinal and 0.9° of lateral cyclic and 1.2° of pitch attitude up to 118 kt, and with the DLR Bo 105-S123 with the collective 0.5–1.1° higher and the main rotor torque within −9 to +2 %. The blades' first flap, lag and torsion frequencies are 1.11, 0.67 and 3.80 per revolution against 1.12, 0.66 and 3.79 from NASA's analysis, and the lag damping follows the full-scale rotor's measured decay (−1.01 against −0.97 s^{-1} at $C_T/\sigma = 0.070$). At 80 kt the roll response to lateral cyclic is within 1.1 dB (rms) of DLR's identified transfer function from 1 to 12 rad/s, including the lag/air resonance near 15 rad/s, and the rate responses to the four AGARD verification doublets have a mean on-axis rms error of 2.6 deg/s. Deficiencies are quantified: the isolated rotor in hover needs 3–11 % less power than the full-scale test at the same thrust and 2.6–3.1° less collective, the tail rotor's pitch is 1.4–2.3° above DLR's through the transition from 20 to 40 kt and −1.4° from it in the hover, several cross-coupling derivatives at 80 kt lie outside the spread of six identifications from the same flights, the pitch and roll damping derivatives are larger than any of them, and in the longitudinal and collective doublets the free model drifts away from the aircraft after about ten seconds.

1 Introduction

The Bo 105 distributed for FlightGear flies on YASim, whose helicopter model builds the rotor from a short list of geometric parameters and tuning factors. For most helicopters that is a reasonable compromise; for the Bo 105 it misses what made the aircraft famous. Its titanium hub carries the blades without flapping or lead-lag hinges: the blades bend instead, their first flap frequency 1.12 times the rotor speed and their first lag frequency 0.66 [2]. The flap frequency gives a hub moment three to four times that of an articulated rotor of the same size, hence the quick, crisp control response and the aerobatic reputation; the low lag frequency puts a regressing lag mode near 15 rad/s, where it couples with the body's roll (air resonance) and has to be damped by the blades' own aerodynamics and structure, since there are no lag dampers.

The data to build a physical model exist in public. Boeing Vertol validated its C-81 rotorcraft analysis against MBB's flight tests of the Bo 105 in 1975 and published the complete input deck, including the blade's mass and stiffness distribution [1]; it is the Bo 105's equivalent of the UH-60's GenHel documentation. The DLR's research Bo 105, S123, was the subject of a coordinated system identification by six organisations under AGARD [7, 8]. NASA tested the full-scale rotor in hover and in the 40×80 ft wind tunnel [3, 4, 2, 5], and the DLR documented its engine and governor model [9].

This report describes a JSBSim [16] model built on that material, after a UH-60A model built the same way, and how well it reproduces the tests. All of the flight model, including the blades' integration, the wake, the engines and the governors, is JSBSim XML; the FlightGear side only animates and instruments it. All result tables, figures and the numbers quoted from them are written by the evaluation scripts.

2 Reference data

Table 1 lists the sources; most are scans, and every curve used was digitised from them. A few points of data reduction matter for the comparisons. *(i)* The DLR trims are those Padfield plots against his Helisim model; their weight, 2200 kg, is given there, the atmosphere is not, and sea level standard is assumed. The pitch harmonics are in Helisim's convention, the collective at the shaft. *(ii)* The MBB trims are those of the V-4 prototype's flight 108 as TR-75-50 plots them against C-81: 4500 lb, 5000 ft density altitude, the CG 3.1 in forward of the rotor reference axis; collective at 0.7 R and cyclic as swashplate angles. *(iii)* The six identifications of AR-280 used the same flight data and differ widely: the roll damping L_p ranges from −2.0 to −8.8 s^{-1}; their structures differ (equivalent time delays, terms eliminated). The comparison here asks whether the model lies within that spread. Two printed values (N_{δ_r} of Glasgow and NAE) are ten times the others' and probably misprinted. *(iv)* The verification doublets of AR-280 (figs. 6.2.17 and 6.2.20) were digitised on the figures' common time axis; each of the four runs is replayed with the recorded increments of all four controls. *(v)* NASA's rotor tests ran on the Rotor Test Apparatus in the test section of the 40×80 ft tunnel, the shaft tilted 10° forward, the first harmonic flapping nulled with cyclic.

Table 1: Sources and what they contribute.

Source	Content used
USAAMRDL TR 75-50 [1]	C-81 input deck: geometry, rigging, NACA 23012 tables, blade mass and stiffness (Y-71 beam model), fuselage wind tunnel data; MBB's V-4 trim flights (fig. 16)
NASA TM-103867 [2]	blade frequencies tuned to MBB's measurements, section axes, control system stiffness
NASA TM-85892, TM-85990 [3, 4]	full-scale rotor in hover: thrust, power, lag damping against thrust
NASA TM-110356 [5]	full-scale rotor in the wind tunnel; hub drag tare
NASA CR-3144 [6]	geometry, loadings, stations
AGARD AR-280 [7]	DLR Bo 105-S123: loading and inertias; derivatives and modes at 80 kt from six identifications; verification doublets
15th ERF, paper 66 [8]	identified roll transfer function at 80 kt (4th/7th order, with the lag mode)
23rd ERF, paper 91 [9]	250-C20B engine and governor structure; collective 3-2-1-1 in hover with rotor speed and torque
Padfield [10], via [11]	DLR S123 trims against airspeed (figs. 4.6, 4.10, 4.11)
EASA TCDS R.011 [12]	engine ratings, speeds and limits
Flight manual CB-5/CBS-5 [13]	stations, loading, cockpit, instrument markings, warning lights
METU thesis [11]	Peters–He finite-state wake equations
NASA TN D-234, NACA TR 1319 [14, 15]	ground effect in forward flight; ring-vortex wake

3 The model

3.1 Implementation

A Python script writes the model from the transcribed tables (Table 2). JSBSim evaluates aerodynamic functions on demand, so the function graph must be free of cycles; every quantity that depends on its own past, the blades' modal coordinates and their rates, the wake states, the rotor speed, the tail rotor's teeter and inflow and the engines' states, therefore lives in flight-control-system channels, which JSBSim runs once per frame in a fixed order and which can write properties back. These channels integrate the states explicitly at 960 Hz, the rate FlightGear runs the model at: 2.65° of rotor azimuth per step, about 0.08 ms of computation. The rate is set by the rotor's air resonance: at 240 and 480 Hz the explicit couplings between the lag modes and the airframe left that mode under-damped or unstable; from 960 Hz on it no longer changes (Section 6). The airframe's loads, which change slowly against the rotors', are worked out in a channel run every fourth frame, at 240 Hz; so are the engines' gas generators, governors and temperatures, the fuel and the outputs to FlightGear, while the rotor speed, the drive train and the power turbines' torque, which damps the drive train mode, stay at 960 Hz. The results here are the same to their printed precision, but for three entries of the doublet and engine tables, which move by one in their last digit. In FlightGear 2024 every property read takes a lock, which makes the reads (about 5 600 per step) the model's cost there, about five times the offline harness's; the segment equations are written to read few, a segment's velocity, for one, from four coefficients per blade, since it is linear in the segment's radius and mode shapes, and a function that only one other reads is written into it. The same XML runs in FlightGear and in the offline harness used here.

Table 2: The generated model: JSBSim system files, their content and the number of named aerodynamic functions or control-system components.

File	Content	n
`aero`	the aerodynamic functions: rotor segments, wake, tail rotor; the six force and moment axes	985
`airframe`	fuselage, hub and skid drag, stabiliser and end plates, fin blockage, at 240 Hz	146
`rotor`	blade modes, rotor azimuth, Peters–He wake states, tail rotor teeter and inflow, torsion	122
`drive`	engines, fuel control and governors, rotor speed, freewheels, rotor brake, fuel	69
`controls`	pilot controls, rigging, boost servos	18
`outputs`	properties for the 3-D model and the instruments	50

3.2 Main rotor

Each blade is a beam cantilevered from the shaft, with the hub's 2.5° of precone built in. Its first flap and first lag bending modes come from the Y-71 beam model of TR-75-50 (23 stations with their mass, pitch inertia and flap, lag and torsion stiffness), solved as Euler–Bernoulli elements with the centrifugal tension's stiffening and, in lag, the centrifugal softening. Table 4 as it stands gives a lag frequency of 0.76 per revolution; MBB's measured value, which NASA's analysis was tuned to [2], is 0.66, and the root's in-plane stiffness is scaled (by 0.19) to give it. The torsion stiffness is scaled likewise to NASA's 3.79 per revolution (Y-71 had 3.18). The modal equations carry the aerodynamic forcing, the hub's specific force and the airframe's angular rates and accelerations, the centrifugal force on the coned blade, the Coriolis coupling between flap and lag through the blade's coning and its own bending, and the structural flap–lag coupling outboard of the pitch bearing, where the blade's principal axes turn with its pitch. Structural damping is 1 % of critical in flap [1]; in lag 1.7 %, fitted to the full-scale rotor's decay (Section 5.3). The first torsion mode is quasi-static: the aerodynamic moment about the elastic axis, the inertial forces at the section's centre of gravity aft of it and the propeller moments, against the torsion and control system stiffness. The modal accelerations take a second-order explicit step.

Eight segments of equal annulus area from 0.2 R outboard see the air velocity at their mid-point from the full kinematics: the aircraft's translation and rotation, the rotor speed, the blades' bending velocities and the induced velocity. Lift and drag come from the C-81 NACA 23012 tables against angle of attack over ±180° and Mach number; the tip loss factor is 0.97, as assumed for the UH-60. The hub loads are summed from the blades' aerodynamic and inertial forces, the latter for their motion relative to the airframe, whose mass JSBSim carries rigidly; the hub moment of the hingeless rotor is therefore the blades' root bending, not a hinge offset's lever. On a soft in-plane rotor the first lag mode carries most of the blades' inertia about the shaft, so the rotor speed and the four lag accelerations are solved together in each step, as in the DLR's

own simulation [9].

3.3 Inflow and wake

The induced velocity comes from the Peters–He generalised dynamic wake [11] with three axisymmetric states and the first harmonic pair, as in the UH-60 model. Ground effect scales the induced velocity: in hover by $(1 + 0.3(R/Z)^2)^{-1/2}$ with Z the hub height, the UH-60 operator's manual hover chart's fit (no Bo 105 hover chart is public), carried into forward flight with the interference ratios of TN D-234 [14]. The main rotor's downwash at the fuselage, the horizontal stabiliser, its end plates and the tail rotor comes from a skewed cylinder of ring vortices leaving the tip-path plane [15], averaged over each surface and tabulated against the wake's skew and direction and the disc's longitudinal tilt.

3.4 Tail rotor

The two-bladed teetering tail rotor sits on the left of the fin and pushes the tail to the right; it is built like the main rotor, as the UH-60 model's. The two blades teeter on the hub without a spring or precone [1]: blade 0 flaps by β towards the thrust side, blade 1 by $-\beta$, and the teeter's equation is the difference of the two blades' flap equations about the centre, with their aerodynamic moments, the centrifugal stiffness, the gyroscopic terms of the body's rates and the hub's angular acceleration. Each blade's pitch is the pedals' less $\tan\delta_3$ times its own flapping ($\delta_3 = 45°$), so the pitch–flap coupling acts on the teeter, which a closed form leaves out. Six segments of equal annulus area from $0.2\,R$ (the deck gives the hub no extent) see the air at the hub, including the main rotor's ring wake, plus the blade's own motion and a uniform dynamic inflow (Pitt–Peters momentum over the effective disc B^2A, $B = 0.94$ from Prandtl's formula at the hover thrust), which reverses with the thrust and is not limited in the vortex ring state. The section is the NACA 0012 of the C-81 deck's own equations for this tail rotor (cards 32A–E): the lift slope and maximum lift against Mach number, the maximum lift in reversed flow, the drag polynomial and the drag divergence, tabulated over ±180°, with a flat plate past the stall. The card prints the drag polynomial's quadratic coefficient as 0.00278 per deg^2, which would give the section a drag coefficient of 0.07 at 5°; it is taken as 0.000278 (0.010 at 5°). The thrust, the in-plane forces and the blades' moments on the hub, the drive's torque among them, come from the blades. Its shaft's 4° toe is left out. It turns 5.25 times as fast as the main rotor, 14° of azimuth per step. Bailey's closed form, as in GenHel, can be built instead; Section 5 compares the two in trim.

Pushing, the rotor draws its air past the fin, and with its thrust reversed blows its wake onto it. The fin's own forces are in the fuselage data, measured without a tail rotor, so the fin blockage is added as in the UH-60 model: a ring-vortex wake of the tail rotor, as the main rotor's [15], gives its induced air velocity over four spanwise strips of the 3-D model's fin (0.38 m^2, its leading edge swept 38°, a third of the radius from the disc; Padfield's 0.805 m^2 [10] matches the fin and the end plates together) against the wake's skew and direction, and each strip adds the difference of its lift and drag with and without that velocity, in the local flow with the main rotor's wake. In the hover the fin's upper strips see 0.5–0.7 of the disc's inflow, crossing them sideways, and the fin pushes against the thrust with 4 lb, 1.3 % of it.

3.5 Airframe

The fuselage forces and moments come from the C-81 deck's quarter-scale wind tunnel coefficients, fin included, against angle of attack and sideslip in the local flow with the rotor's wake; flying backwards they are mirrored front to back. The rotor hub, shaft and controls add 4.2 ft^2 of drag, the full-scale rotor's tare with the blades off [5], and the skid gear the cross-flow drag of its tubes. The horizontal stabiliser is a strip with its own lift and drag over ±180° (the C-81 section's slope at its aspect ratio of 8.09, stall at 13°), lifting from +4°, Padfield's zero-lift angle for the S123 [10]; the deck's 0° gave a speed stability M_u above all six identifications at 80 kt and a longitudinal cyclic 23–40 % steeper against speed than both flight tests'. Its two end plates are low aspect ratio vertical plates.

3.6 Flight controls

The rigging from the pilot's controls to the swashplate and the tail rotor is the C-81 deck's, with the swashplate's 10° control phasing; the basic Bo 105 CBS has no mixing and no stability augmentation. The hydraulic boost servos are second-order systems with the UH-60's primary servo response, for want of Bo 105 data.

3.7 Engines, drive and mass

Each Allison 250-C20B is a nonlinear model of the same structure as the UH-60's T700: the fuel control holds a gas generator speed reference within acceleration and deceleration schedules, the gas generator follows with a speed-dependent lag, and the power turbine's torque depends on both speeds, shaped to the type certificate's ratings [12]: 298.8 kW per engine at 100 % torque, 86 % for take-off and 110 % at 105 % gas generator speed. The power turbine governor, after the DLR's description [9], is proportional on the rotor speed error with a feed forward from the collective, a lag and a slow reset (20 s); its gain, lag and feed forward were fitted to the DLR's hover collective 3-2-1-1 (Section 5.7). Freewheel clutches connect the engines; the rotor speed is a degree of freedom with the hub's and the blades' inertia, the lag modes carrying the blades' motion relative to the hub. The gearbox absorbs 2.5 % of the torque and the accessories 8 kW (both estimates). The default loading is the DLR S123's 2200 kg with its inertias [7]; the loading stations are the flight manual's [13]. The skids are sliding contacts at the 3-D model's skid tubes. Table 3 lists the principal data.

4 Method

The evaluation drives JSBSim from Python with the same files as FlightGear, at 960 Hz. A trim solves for the four controls and the attitudes that zero the six accelerations averaged over two rotor revolutions (Hann-weighted), with the airframe held while the rotor and wake settle. The programme:

- *level-flight trims*: the DLR S123 (2200 kg, sea level) from hover to 120 kt; MBB's V-4 (4500 lb, 5000 ft, CG 3.1 in forward) from hover to 118 kt;

Table 3: Principal data. Sources: TR [1], PT [2], AR [7], HF [6], EA [12], HG [9].

Quantity	Value	Source
Gross weight (default, DLR S123)	2200 kg (4850 lb)	AR
Main rotor radius, blades, chord	16.11 ft (4.91 m), 4, 0.887 ft	TR
Solidity, twist, precone	0.070, −8°, 2.5°	TR, PT
Rotor speed, 100 %	44.4 rad/s (424 rpm)	EA, HG
Shaft tilt, control phasing	3° forward, 10°	TR
Section; segments	NACA 23012 (C-81); 8, from 0.2 R	TR
Blade modes, per rev	flap 1.11, lag 0.67, torsion 3.80	TR, PT
Blade weight, flap inertia	115.0 lb, 162.1 slug ft^2	TR
Tail rotor: radius, blades, chord	3.115 ft, 2 (teetering), 0.588 ft	TR, AR
Tail rotor: gear ratio, δ_3	5.25, 45°	TR, HF
Tail rotor: section; segments	NACA 0012 (C-81 equations); 6, from 0.2 R	TR
Stabiliser area, aspect ratio	8.71 ft^2, 8.09, end plates	TR, EA
Hub, tail rotor hub (FS, BL, WL)	(98.4, 0, 61.2), (335, -12.5, 68.7) in	TR, HF
Default CG (FS, WL)	(98.4, 6.9) in	HF
Inertias I_x, I_y, I_z, I_{xz}	1433, 4973, 4099, 660 kg m^2	AR
Engines	2 × Allison 250-C20B, 298.8 kW at 100 % torque	EA
Drive: rotor inertia	693 slug ft^2 (blades 648)	TR, est.
Simulation rate	960 Hz	

- *the isolated rotor in hover*: the whole aircraft held far from the ground at 425 rpm, sea level, its cyclic nulling the first harmonic flapping, at the seven collectives of the full-scale test;
- *blade dynamics*: the bending frequencies; the lag damping in hover, the airframe and rotor speed held, one blade's lag kicked and a damped sinusoid fitted to its decay, at six collectives;
- *derivatives and modes* at 80 kt, 3000 ft, the S123's loading of the identification flights: central differences of the forces and moments, rotor included, with the airframe held and the rotor running; the modes of the resulting 8-state model;
- *roll frequency response* at 80 kt: the free aircraft flown with a 60 s logarithmic lateral cyclic sweep of 1.5 % from 0.7 to 30 rad/s, pitch attitude held loosely through the longitudinal cyclic, the rotor speed governed;
- *verification doublets* at 80 kt: the recorded increments of all four controls of each AGARD run, applied open loop from trim, no stability augmentation (the S123 flew them without);
- *collective 3-2-1-1 in hover* (2200 kg, 1000 ft): the recorded collective applied with the rotor speed free, pitch and roll held by a cyclic feedback on attitude and rate (the DLR's run was "with attitude stabilisation").

Table 4: Level-flight trim, model minus DLR test (deg); torque as model/test −1.

kt	θ_0	θ_{1s}	θ_{1c}	θ_{0T}	θ	Q, %
0	+0.67	−1.34	−0.96	−1.40	−0.10	–
10	+1.05	−1.35	−0.82	−0.35	+0.16	−1
20	+1.07	−0.31	−1.42	+1.44	−0.81	+1
30	+1.00	−0.26	−1.19	+2.34	−1.12	+2
40	+0.83	−0.19	−0.73	+2.04	+1.25	+2
50	+0.73	−0.52	−0.27	+1.06	+1.20	+1
60	+0.71	−0.55	−0.05	+0.45	+0.90	−4
70	+0.55	−0.59	+0.02	+0.07	+1.12	−4
80	+0.52	−0.59	+0.12	−0.24	+0.41	−8
90	+0.50	−0.56	+0.14	−0.37	+0.31	−9
100	+0.69	−0.65	+0.21	−0.31	–	−7

Table 5: Level-flight trim, model minus MBB's test (deg), and the model's total power.

kt	$\theta_{.7}$	B_{1s}	A_{1s}	θ	hp
0	+0.10	−0.58	+0.59	−0.86	496
20	−0.05	−0.75	+0.65	+0.07	433
33	−0.51	−1.20	+0.94	+0.07	360
44	−0.95	−1.12	+0.37	+0.71	324
55	−0.04	−0.42	+0.29	+0.46	307
66	−0.26	−0.22	+0.27	+0.17	309
78	−0.67	−0.20	+0.28	−0.58	331
88	−0.64	−0.04	+0.13	−0.99	367
98	−0.17	−0.04	+0.04	−1.06	419
108	−0.02	−0.07	−0.23	−0.41	489
118	+0.40	+0.07	−0.36	−1.22	583

5 Results

5.1 Level-flight trim

Against MBB's flight test (Fig. 2, Table 5) the model's collective at 0.7 R is within 1.0° at every speed, its longitudinal cyclic within 1.2° (0.4° from 55 kt up), its lateral cyclic within 0.9° and its pitch attitude within 1.2°. The power reaches its minimum of 307 hp near 55 kt. The largest differences are in the transition, 33 to 44 kt, where the test needs about 1° more collective and 1.2° more forward cyclic than the model.

Against the DLR S123 (Fig. 1, Table 4) the model needs 0.5–1.1° more collective than the test throughout, and 0.2–1.4° more forward cyclic, 0.6° at most from 60 kt up. The lateral cyclic is within −1.4 to +0.2°, the largest differences at 10–30 kt, where the test's lateral cyclic peaks with the onset of translational lift. The main rotor torque is within 4 % up to 60 kt and falls to −9 % of the test's at 90 kt; at the lowest test speed, 4 kt, the test reads 8.92 kN m and the model's hover 7.87. The tail rotor pitch (Fig. 1d) is −1.4° from the test's in the hover, 1.4–2.3° above it from 20 to 40 kt and within 0.5° from 60 kt up. Bailey's closed form, which the model used before, is within +0.4° in the hover but 3.0–3.6° above the test through the transition and within 1.1° from 60 kt up: the blade elements make more thrust per degree and less of the hover's pitch, and follow the test's fall through the transition more closely. Both need 34 hp or so in the hover. In the test the tail rotor's pitch drops by 4° between 12 and 23 kt, which points to the fin and the tail rotor leaving the main rotor's wake earlier than the ring wake has it. The fin blockage is small, as the fin

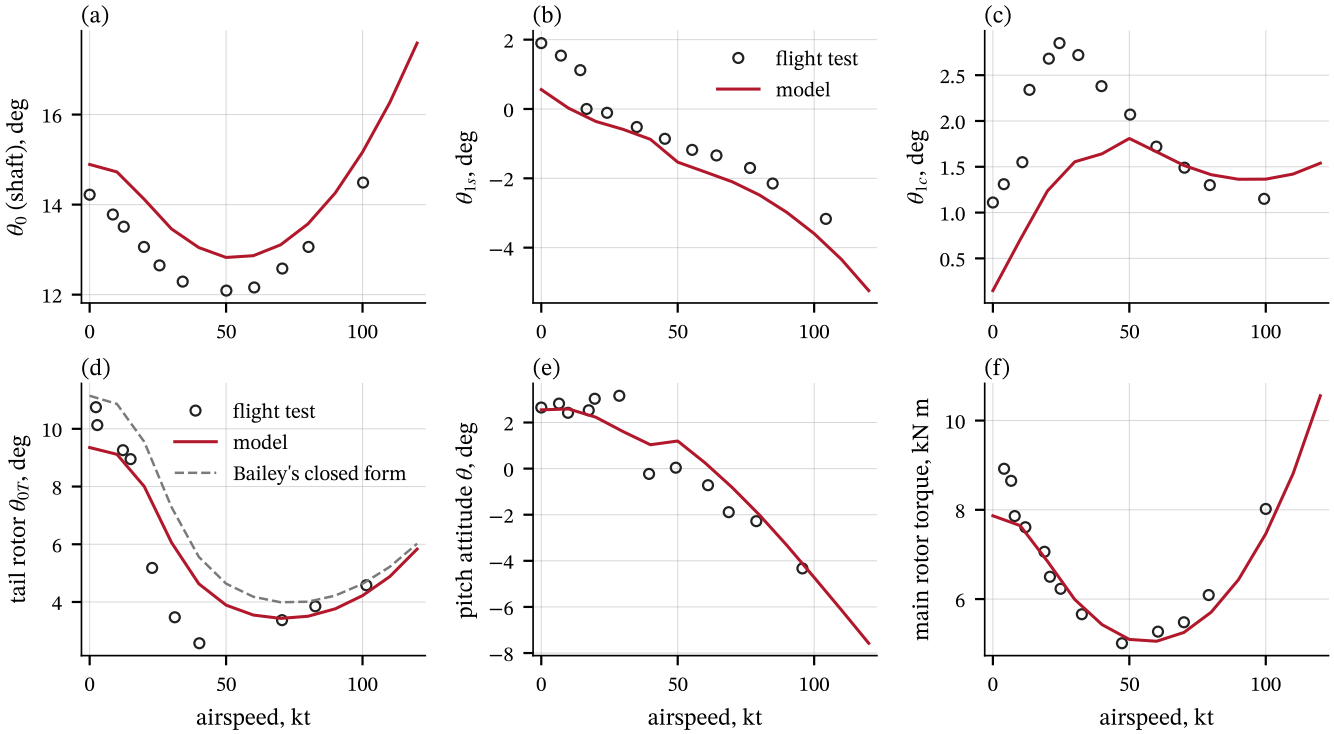


Figure 1: Level-flight trim against the DLR Bo 105-S123 as published by Padfield [10]: 2200 kg, sea level. Blade pitch harmonics in Helisim's convention ($\theta_{1s} < 0$ forward, $\theta_{1c} > 0$ left), collective at the shaft; main rotor torque.

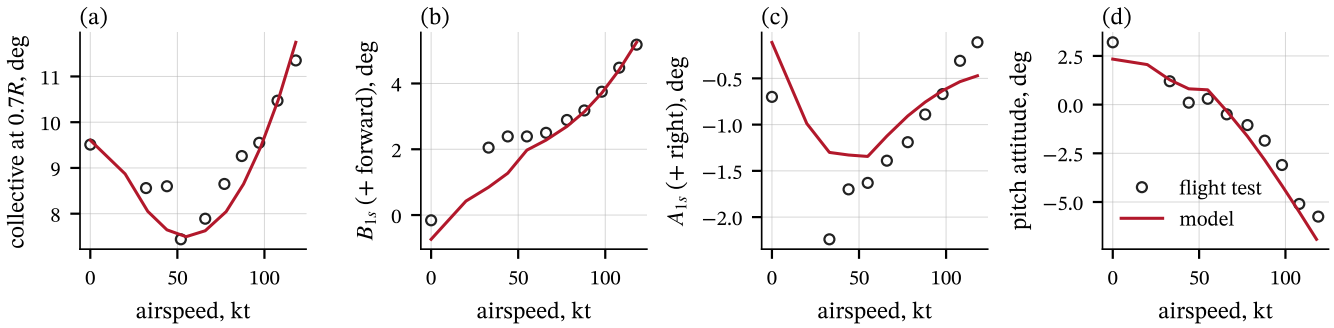


Figure 2: Level-flight trim against MBB's V-4 flight 108 as plotted in TR-75-50 fig. 16: 4500 lb, 5000 ft density altitude, CG 3.1 in forward.

stands in the pusher's inflow rather than its wake: 1.3 % of the tail rotor's thrust in the hover, 4 % at 30 kt, where the inflow's sidewash on the fin is largest against the airspeed, and 2 % or less from 60 kt up. It raises the pitch by 0.08° in the hover and 0.2° at most, so it explains neither difference. The pitch attitude is within 1.3°. The two tests together suggest a model that is right in shape and slightly heavy in collective for the DLR aircraft, whose exact atmosphere is not published. The model trims with −1.3 to −3.3° of roll, left side down; no test value is published.

5.2 The isolated rotor in hover

Here the model departs most clearly from a test (Fig. 3, Table 6). At the same collective it makes more thrust: the curves are offset by 2.6–3.1° of collective at equal C_T/σ, with nearly the same slope. At the test's thrust it needs 3–11 % less power, and its figure of merit peaks at 0.73 against the test's 0.66. The steady torsion moment at 0.34 R has the test's trend but −333 against −480 in lb at $C_T/\sigma = 0.07$. In the flight trims the same rotor needs *more* collective than the DLR aircraft (above), so a collective offset in the model's rigging would not explain both. Possible causes, none verified: the tunnel

Table 6: The isolated rotor in hover: thrust at the test's collectives; power and figure of merit at the test's thrust.

$\theta_{.7}$	C_T/σ		$10^3 C_P/\sigma$ at test C_T		FM	
deg	test	model	model	test	model	test
3.3	0.0000	0.0150	–	1.09	–	–
6.7	0.0199	0.0443	–	1.84	–	0.285
8.9	0.0411	0.0668	3.00	3.09	0.520	0.504
11.2	0.0620	0.0931	4.46	4.79	0.647	0.602
13.0	0.0788	0.1148	5.97	6.41	0.693	0.645
14.9	0.1047	0.1368	8.78	9.57	0.722	0.662
16.5	0.1194	0.1506	10.75	12.14	0.718	0.635

test section in which the rotor hovered recirculates part of the wake, which raises the inflow and the power at a given thrust; the test's collective was measured at the blade's root, inboard of the control system's and the blade's elastic twist under the torsion moment, which the model's quasi-static torsion mode represents with a lower moment; and the model's section drag is that of the smooth C-81 tables, without the increment for a tripped boundary layer used for the UH-60. The tunnel's recirculation, if it is the cause, would put the model closer to

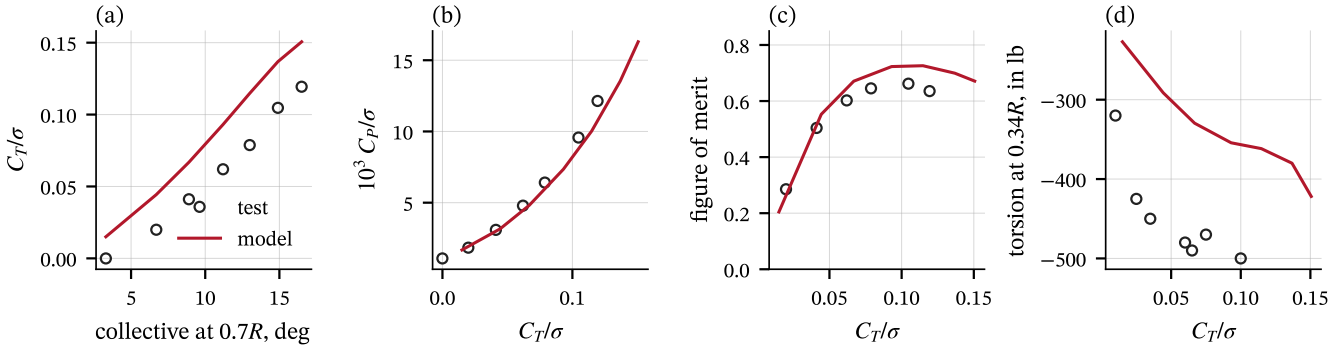


Figure 3: The isolated rotor in hover at 425 rpm against NASA's full-scale test (TM-85892; torsion TM-103867): (a) thrust against collective at 0.7 R, (b) power against thrust, (c) figure of merit, (d) steady blade torsion moment at 0.34 R.

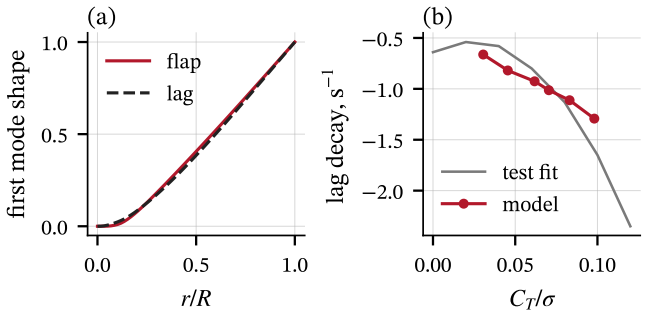


Figure 4: (a) The first flap and lag modes. (b) Decay of the regressing lag mode in hover, 425 rpm, against the curve fit through the full-scale rotor's measurements (TM-85892 fig. 6).

Table 7: Blade frequencies per revolution at 425 rpm: the beam model of TR-75-50's table 4 (torsion stiffness scaled), the model (lag stiffness tuned), Boeing's Y-71 results and NASA's comprehensive analysis tuned to MBB's measurements.

Mode	beam	model	Y-71 [1]	analysis [2]
Flap 1	1.109	1.109	1.125	1.12
Lag 1	0.763	0.674	0.732	0.66
Torsion 1	3.790	3.798	3.176	3.79
Flap 2	2.808	–	2.780	2.78
Lag 2	4.584	–	4.518	4.33
Flap 3	5.007	–	5.007	5.02

Table 8: Lag damping in hover: decay coefficient of one blade's lag after a kick, and the full-scale rotor's (curve fit, TM-85892 fig. 6).

C_T/σ	decay, s^{-1}		ν_ζ
	model	test fit	/rev
0.0305	−0.66	−0.56	0.671
0.0455	−0.82	−0.64	0.665
0.0618	−0.93	−0.83	0.662
0.0704	−1.01	−0.97	0.661
0.0831	−1.11	−1.21	0.659
0.0981	−1.29	−1.60	0.655

free-air hover than the test.

5.3 Blade dynamics

The first flap, lag and torsion frequencies (Table 7) are 1.11, 0.67 and 3.80 per revolution, against 1.12, 0.66 and 3.79 from NASA's analysis; the higher modes of the beam model agree with Boeing's to within 2 %, which checks the transcription of the structural data. The lag damping (Fig. 4b, Table 8) is fitted at one point, $C_T/\sigma \approx 0.07$, with a structural damping ratio of 1.7 % (with 1 % the decay there was −0.77 s^{-1} against −0.97); the rest follows from the blades' aerodynamics. The model's decay grows with thrust as the test's does, but less steeply: −0.66 against −0.56 s^{-1} at $C_T/\sigma = 0.031$ and −1.29 against −1.60 at 0.098. The fitted lag frequency is 0.66 per revolution, on the 0.66 of the measurements.

5.4 Derivatives and modes at 80 kt

Of the 36 derivatives that at least four of the six identifications include, 17 of the model's lie within their range (Table 9; the model's trim at 80 kt is −1.6° of pitch and −1.7° of roll). Within it are the speed and heave derivatives X_u, Z_u and Z_{δ_c}, the angle-of-attack stability M_w, the dihedral L_v and the yaw damping N_r. The cyclic control powers lie above the range, $M_{\delta_e} = 0.140$ against 0.050 to 0.098 and $L_{\delta_a} = 0.202$ against 0.053 to 0.185, and the damping derivatives further still: $M_q = -6.755$ against −4.493 to −1.441, $L_p = -10.586$ against −8.779 to −1.995. Their ratios, the steady rate per % of cyclic, are close to the DLR's: 1.9 against 2.2 rad/s per 100 % in roll and 2.1 against 2.7 in pitch. A quasi-steady derivative lets the rotor settle at each perturbation, which a 6-degree-of-freedom identification of an aircraft whose rotor lags the body cannot; the identifications themselves differ by a factor of four in L_p. The directional stability $N_v = 0.178$ is above the range (0.051 to 0.117). The clearest disagreements are in the couplings: the roll due to longitudinal cyclic L_{δ_e} (−0.071 against −0.005 to 0.073), the roll due to collective L_{δ_c} (−0.017 against 0.004 to 0.058), the yaw due to roll rate N_p (0.033 against −1.958 to −0.466) and the pitch due to collective, $M_{\delta_c} = 0.090$ (0.032 to 0.073). The C-81 analysis had the same difficulty with the off-axis response [1]. The modes (Table 10) follow: the Dutch roll [0.20, 2.75] has its damping within the identified 0.13–0.22 and its frequency 6 % above the highest of 2.17–2.60; the phugoid [−0.14, 0.29] is unstable, as identified (−0.07 to −0.36); roll and short-period pitch combine into a critically damped pair at 8.96 rad/s, where the identifications find roll subsidences of 2.4–8.5 s^{-1} and pitch roots of 1.4–6.0.

5.5 Roll frequency response

The frequency response (Fig. 5) is the test that sees the rotor's dynamics directly. From 1 to 12 rad/s the model's magnitude is within 1.1 dB (rms, 1.8 at most) of the identified transfer

Table 9: Derivatives at 80 kt, 3000 ft, 2200 kg: the model, the DLR's identification, and the range of the six identifications of AGARD AR-280 (AFDD, CERT, DLR, Glasgow, NAE, NLR). Body axes at the CG, SI units, controls in % of travel; ↑↓: above or below the range.

	model	DLR	six identifications		model	DLR	six identifications
X_u	−0.047	−0.059	−0.059 .. −0.032	Y_r	0.305↓	1.332	0.807 .. 1.332
X_w	0.024	0.036	−0.061 .. 0.036	Y_{δ_a}	0.018	0.003	−0.001 .. 0.076
X_q	0.828↓	–	1.206 .. 2.548	Y_{δ_r}	−0.026	−0.011	−0.071 .. 0.015
X_{δ_e}	−0.025↑	−0.028	−0.072 .. −0.027	L_v	−0.264	−0.271	−0.271 .. −0.038
Z_u	−0.012	0.014	−0.088 .. 0.246	L_p	−10.586↓	−8.501	−8.779 .. −1.995
Z_w	−0.758↑	−0.998	−1.187 .. −0.791	L_q	3.819	3.037	0.558 .. 4.454
Z_q	−0.140↓	5.012	5.012 .. 7.011	L_r	−0.047↓	0.410	0.410 .. 1.321
Z_{δ_c}	−0.268	−0.349	−0.388 .. −0.242	L_{δ_a}	0.202↑	0.185	0.053 .. 0.185
Z_{δ_e}	−0.054	−0.303	−0.303 .. −0.008	L_{δ_e}	−0.071↓	0.024	−0.005 .. 0.073
M_u	0.031↑	0.029	0.008 .. 0.029	L_{δ_c}	−0.017↓	0.032	0.004 .. 0.058
M_w	0.086	0.053	0.019 .. 0.356	L_{δ_r}	−0.053↓	−0.028	−0.028 .. −0.010
M_p	−1.215	−0.419	−1.414 .. −0.419	N_v	0.178↑	0.117	0.051 .. 0.117
M_q	−6.755↓	−3.496	−4.493 .. −1.441	N_p	0.033↑	−1.057	−1.958 .. −0.466
M_{δ_c}	0.090↑	0.057	0.032 .. 0.073	N_q	0.499	0.809	0.491 .. 5.432
M_{δ_e}	0.140↑	0.093	0.050 .. 0.098	N_r	−1.093	−0.858	−1.116 .. −0.699
M_{δ_a}	0.039↑	−0.009	−0.009 .. 0.011	N_{δ_r}	0.081	0.049	0.043 .. 0.484
Y_v	−0.136	−0.170	−0.279 .. −0.064	N_{δ_a}	−0.000↓	0.026	0.015 .. 0.047
Y_p	−0.704↑	–	−2.993 .. −2.030	N_{δ_c}	0.030↑	–	−0.051 .. −0.036

Table 10: Modes at 80 kt: oscillatory [damping ratio, natural frequency in rad/s], aperiodic $1/T$ (s^{-1}, stable > 0).

Mode	model	identified (AFDD, CERT, DLR, Glasgow, NAE, NLR)
phugoid	[−0.14, 0.29]	[−0.36, 0.30], [−0.17, 0.32], [−0.15, 0.33], [−0.10, 0.35], [−0.14, 0.33], [−0.07, 0.33]
dutch roll	[0.20, 2.75]	[0.22, 2.60], [0.13, 2.51], [0.14, 2.50], [0.16, 2.27], [0.13, 2.58], [0.17, 2.17]
roll / pitch	[1.00, 8.96]	roll 8.32, [0.99, 2.89], 8.49, 5.12, 8.47, 2.38; pitch 6.04, 4.36, 1.98, 4.38, 1.37
pitch (heave)	0.46	0.49, 0.66, 0.60, 0.64, 0.63, 0.71
spiral	0.02	0.03, −0.05, 0.02, −0.01, 0.03, 0.04

Table 11: Verification doublets: rms error of the rate increments (rad/s) over the first 10 s; bold: on axis.

Doublet	p	q	r
longitudinal	0.031	**0.030**	0.032
lateral	**0.039**	0.033	0.045
pedal	0.020	0.018	**0.080**
collective	0.047	**0.034**	0.023

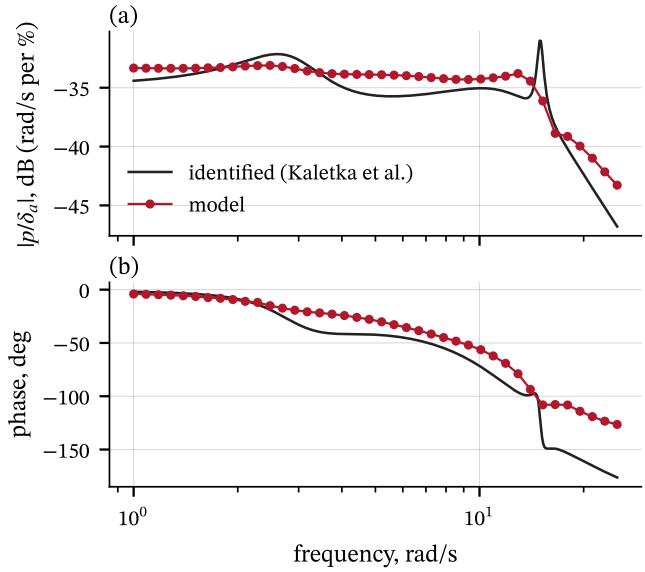


Figure 5: Roll rate per % lateral cyclic at 80 kt: the model's frequency response from a lateral cyclic sweep, and the transfer function the DLR and the AFDD identified from the flight frequency response (Kaletka et al. [8], with the lag/air resonance and the body roll/regressing flap mode).

function and its phase within 11° (rms); the model has a little less phase lag between 3 and 12 rad/s. The lag/air resonance appears at the identified frequency near 15 rad/s with a similar notch and peak. Above it the model's phase lags the identified by about 43° less; the identified function includes a 17 ms time delay, which accounts for 15–25° there.

5.6 Verification doublets

Over the first ten seconds (Fig. 6, Table 11) the on-axis rms error is 2.6 deg/s on average and 2.1 deg/s over all twelve traces. The pitch response to the longitudinal doublet peaks at −7.2 deg/s against −6.9, and the pedal doublet's yaw rate at 14 deg/s against 13, its oscillation, the Dutch roll, in phase with the record. The lateral doublet's roll response is close in its first second and then rings for longer than the aircraft's. The off-axis responses carry the couplings of Section 5.4: the pitch rate after the lateral doublet is about half the record's, and the roll rate after the collective doublet has the wrong sign. After about ten seconds the model drifts away from the aircraft in the longitudinal and collective runs, a pitch and roll divergence that the record does not show although the model receives the pilot's recorded inputs; it is the long-term response of an unaugmented aircraft whose phugoid and speed stability differ slightly from the real one's.

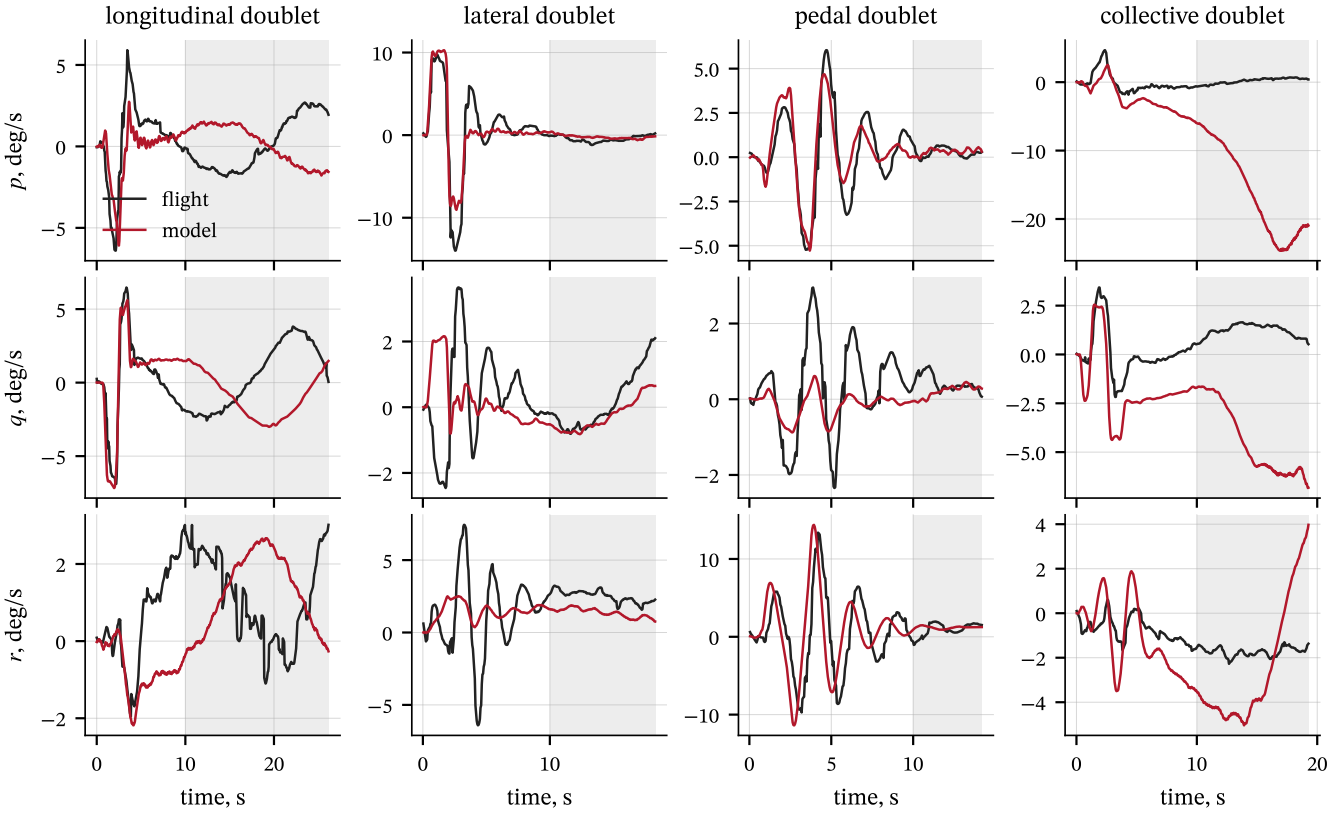


Figure 6: Rate responses to the AGARD verification doublets at 80 kt (AR-280 figs. 6.2.17 and 6.2.20): flight record and model, as increments from the run's quiet window, with the recorded increments of all four controls applied to the model. Shaded: after the 10 s used for the error.

Table 12: Collective 3-2-1-1 in hover: lowest and highest increments in 15 s (yaw rate: 6 s) and the rms error.

	lowest		highest		rms
Increment	test	model	test	model	error
rotor speed, rad/s	−1.32	−1.20	+1.03	+0.79	0.166
shaft torque, kN m	−1.91	−1.40	+1.31	+1.92	0.328
lag, tip/$R \times 10^3$	−2.25	−4.16	+3.33	+3.02	1.2
yaw rate, deg/s (6 s)	−8.84	+0.00	+19.61	+24.67	4.23
vert. sp. force, m/s^2	−3.35	−2.91	+2.43	+2.17	0.467

5.7 Rotor speed and engine response

The governor's gain, lag and collective feed forward were fitted to this record, so it checks the structure rather than predicts (Fig. 7, Table 12). The rotor speed droops by −1.20 rad/s against −1.32 and overshoots by 0.79 against 1.03, an rms error of 0.17 rad/s (0.37 % of the rotor speed). The shaft torque's swing is a little larger than the test's on the way up (1.9 against 1.3 kN m) and smaller on the way down; the vertical specific force follows within 0.47 m/s^2 rms. The blades' collective lag, which the DLR measured and the model computes from its lag modes, follows the record in phase and amplitude. The yaw rate is comparable only in the first seconds, before the pilot's pedal.

6 Corrections found during validation

The changes that the comparisons forced are listed here because each removed a physical or numerical inconsistency rather than tuning a coefficient.

1. *Simulation rate.* The model first ran at 240 Hz. There, and at 480 Hz, the regressing lag mode coupled with the body's roll was under-damped or unstable at 80 kt, through explicit couplings in the step; at 960 Hz its damping no longer changes, and the frequency response shows the identified air resonance. FlightGear runs the model at 960 Hz.
2. *Rotor speed and lag together.* With the rotor speed and the blades' lag integrated one after the other, the lag modes, which carry most of the blades' inertia about the shaft, and the drive exchanged their accelerations a step late. They are solved together.
3. *The blades' inertia counted once.* The flap modes carry the blades' flapping inertia about the shaft's in-plane axes (317 of the rigid blades' 324 slug ft^2); it is taken out of the airframe's roll and pitch inertias, which JSBSim would otherwise count twice.
4. *The hub's acceleration.* The hub is 4.5 ft above the centre of gravity; the blades now feel its specific force, including the airframe's angular acceleration at that arm, as the airframe feels their in-plane forces at the same arm.
5. *A second-order step for the blade modes.* A first-order step turned the flapping's phase by several degrees at 240 Hz, through the half-step lag of the strong aerodynamic damping.
6. *Governor at flat pitch.* The collective feed forward fitted about the hover, held down to flat pitch, left the governor an 11 % droop there; the rotor overshot to 107.5 % after a start. Below 35 % collective it now runs linearly to the gas generator speed the rotor needs at flat pitch, as a compensator cam would; the start peaks at about 101 %, and the hover and forward flight stay on the fitted slope.
7. *The 3-D model's rotor.* The FlightGear model's rotor stands 0.24 m further aft of its nose than the flight man-

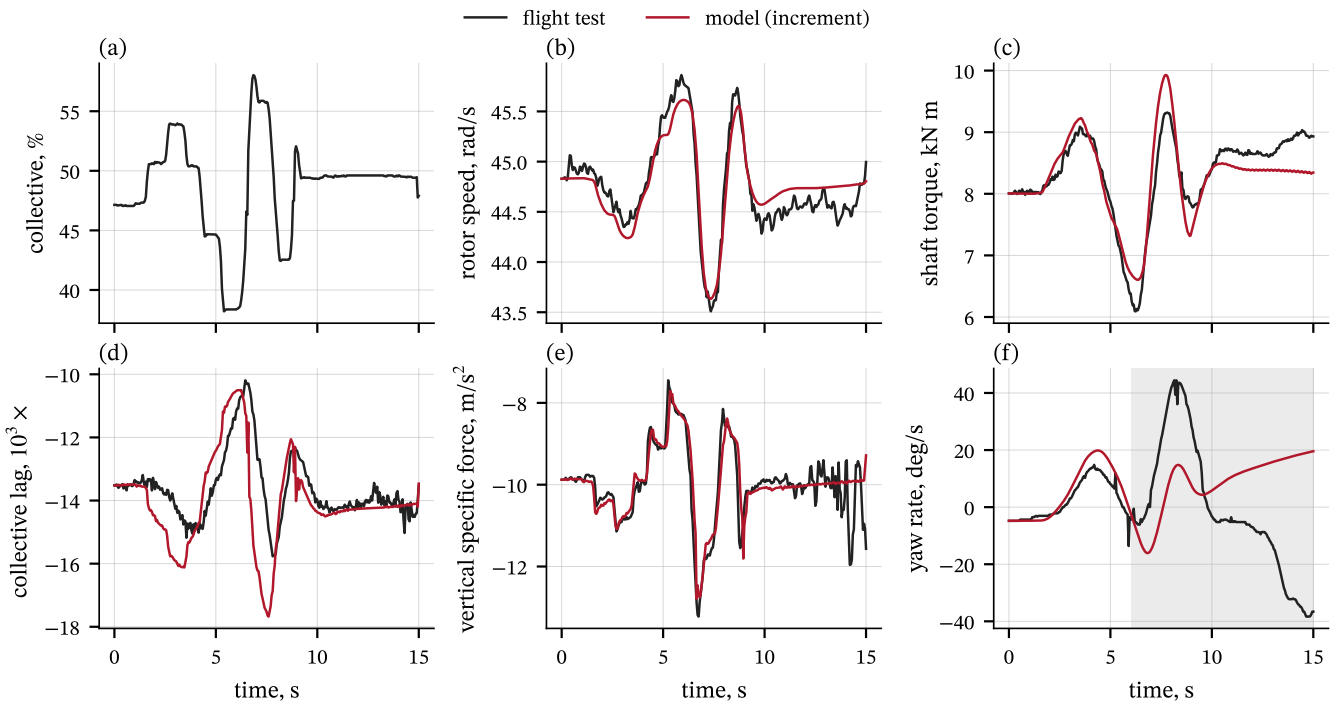


Figure 7: Collective 3-2-1-1 in hover against the DLR's flight test (Hamers and von Grünhagen [9], fig. 6): (a) collective, (b) rotor speed, (c) shaft torque, (d) collective lag (model: mean lag deflection at the tip over R), (e) vertical specific force, (f) yaw rate (the pilot's corrections after 6 s, shaded, are not known). Model as increments on the test's initial value.

Table 13: Summary: ✓ within the test's scatter, ~ partly, × not reproduced.

Trim controls and attitudes against MBB's V-4, hover to 118 kt	✓
Trim against the DLR S123: pitch attitude, torque to 60 kt	✓
Collective and cyclic against the DLR S123 (up to 1.1° and 1.4°)	~
Tail rotor pitch: transition (1.4–2.3° high), hover (−1.4°)	~
Main rotor torque at 80–100 kt (DLR)	~
Isolated rotor: thrust per collective, power per thrust	×
Blade frequencies	✓
Lag damping against thrust (fitted at one point)	~
On-axis derivatives, directional stability, Dutch roll	✓
Damping derivatives L_p, M_q	~
Cross-coupling derivatives	×
Roll frequency response, 1–15 rad/s, air resonance	✓
On-axis doublet responses, first seconds	✓
Off-axis doublet responses; long-term drift	~
Rotor speed after a collective 3-2-1-1 (fitted)	✓

ual's three-view shows; the airframe's stations (skids, end plates, eye point, contacts) are taken from the manual's station diagram, and the visual rotor stays on the 3-D model's mast. The trims, derivatives and responses were rerun after the change and did not move.

7 Assessment

Table 13 summarises. The model reproduces what distinguishes the Bo 105, its blades' dynamics and the quick on-axis response of the hingeless rotor, together with the trims of MBB's prototype. Its open items are the isolated rotor's thrust and power against NASA's hover test, which the flight trims do not confirm and whose cause is open, the tail rotor's pitch through the transition, the cross-couplings, where the model shares the C-81 analysis's difficulty, and the long-term response of the free aircraft. The tail rotor's blade elements have a stall and reversed flow, but no published test checks them at the tail rotor's limits, in strong crosswinds or near its stall, and its section is C-81's equations rather than measured tables. Ground effect uses the UH-60's chart for want of the Bo 105's.

8 Conclusions

- An elastic-blade model of the Bo 105's hingeless rotor, with individual blades, their first flap and lag bending modes and a quasi-static torsion mode, a finite-state dynamic wake, the C-81 airframe and two 250-C20B engines, runs in plain JSBSim, generated from data. It is the flight model of the FlightGear Bo 105.
- The blade frequencies from the published structural data, with the lag stiffness tuned to MBB's measured frequency, and the lag damping fitted at one point, give the roll response at 80 kt with the air resonance where the DLR identified it.
- Against MBB's flight test the trims are within about a degree of every control from hover to 118 kt; against the DLR's aircraft the collective is 0.5–1.1° high and the tail rotor's pitch falls later through the transition than in the test.
- Against NASA's full-scale hover test the rotor makes more thrust per degree of collective and needs less power per unit thrust; the cause is open.
- A blade-element teetering tail rotor with C-81's 0012 equations follows the DLR aircraft's tail rotor pitch through the transition about a degree closer than Bailey's closed form, and is −1.4° from the test in the hover.
- The air resonance sets the simulation rate: an explicit inte-

gration needs 960 Hz for the coupled lag and body modes.
- The cross-coupling derivatives at 80 kt and the free aircraft's long-term drift are documented, not tuned away.

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